

SUMIN KANG

OBJECTIVE

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Junior student in Electrical and Electronic Engineering at Yonsei University, with project experience in imitation learning, multi-agent reinforcement learning, and robot/autonomous driving systems. I am interested in reliable policy learning from imperfect data and its application to practical sequential decision-making problems.

EDUCATION

- **Yonsei University** Mar 2026 - Present
B.S. in Electrical and Electronic Engineering
◦ Admitted as a transfer student (Junior year).
◦ **Relevant Coursework:** Probability and Random Variables
- **Soongsil University** Mar 2020 - Dec 2025
Undergraduate Studies in Artificial Intelligence
◦ GPA: 4.1 / 4.5
◦ **Relevant Coursework:** Autonomous Driving Programming, Machine Learning, Linear Algebra, Data Structures
◦ Academic Excellence Scholarship (Fall 2023 - Spring 2025)

EXPERIENCE

- **NAVER Cloud** Jun 2024 - Feb 2025
AI/Data Intern Seongnam, South Korea
◦ Built a financial statement-based Q&A agent using LangChain, covering data preprocessing, retrieval pipeline construction, prompt design, and deployment.
◦ Evaluated HyperCLOVA X against other LLMs through pilot experiments on task performance, response quality, and practical applicability.

PROJECTS

- **Kaggle Orbit Wars** Jun 2026 - Present
Tools: PyTorch
◦ Develop a multi-agent policy learning pipeline using expert replay data, behavioral cloning, and PPO-based self-play for the Kaggle Orbit Wars environment.
◦ Combine structured planning heuristics with learned target-selection policies, and analyze training behavior using reward, entropy, and outcome-rate metrics.
- **ACT-based Robot Manipulation on SO-ARM101** Apr 2026 - Jun 2026
Tools: PyTorch [Website](#)
◦ Built an imitation learning pipeline for fruit-object pick-and-place tasks using the SO-ARM101 robot arm.
◦ Collected real-world demonstrations through leader-follower teleoperation and trained an ACT-based policy for physical robot manipulation.
- **F1TENTH Autonomous Racing Simulation** Mar 2025 - Jun 2025
Tools: ROS2, Python, C++, Linux [Website](#)
◦ Implemented [TinyLidarNet](#) and integrated it into a ROS2-based F1TENTH simulation pipeline for LiDAR-only end-to-end driving.
◦ Built a ROS2 inference pipeline that converted 1D LiDAR scans into steering and speed commands in simulation.

HONORS AND AWARDS

- **2nd Place, SCON Campus Programming Contest** May 2024
Soongsil University
- **3rd Place, SCON Campus Programming Contest** May 2023
Soongsil University
- **Rank 122nd, ICPC Asia Preliminary Contest** Oct 2023
ICPC

EXTRACURRICULAR ACTIVITIES

- **Roboin, Yonsei University** Apr 2026 - Present
Robotics Club
◦ Work with club members on robotics projects involving hardware setup, control, and robot learning applications.
- **SCCC, Soongsil University** Mar 2023 - Dec 2025
ICPC Study Group
◦ Actively practiced competitive programming and algorithm optimization with team members.

SKILLS

- **Programming:** Python, C/C++, PyTorch, Git, Linux, ROS2